MILITARY SPECIFICATION SHEET

SWITCH ASSEMBLIES, LINEAR MOVEMENT, MANUAL AND REMOTE OPERATIONS, STYLE 5BLS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-S-24187.

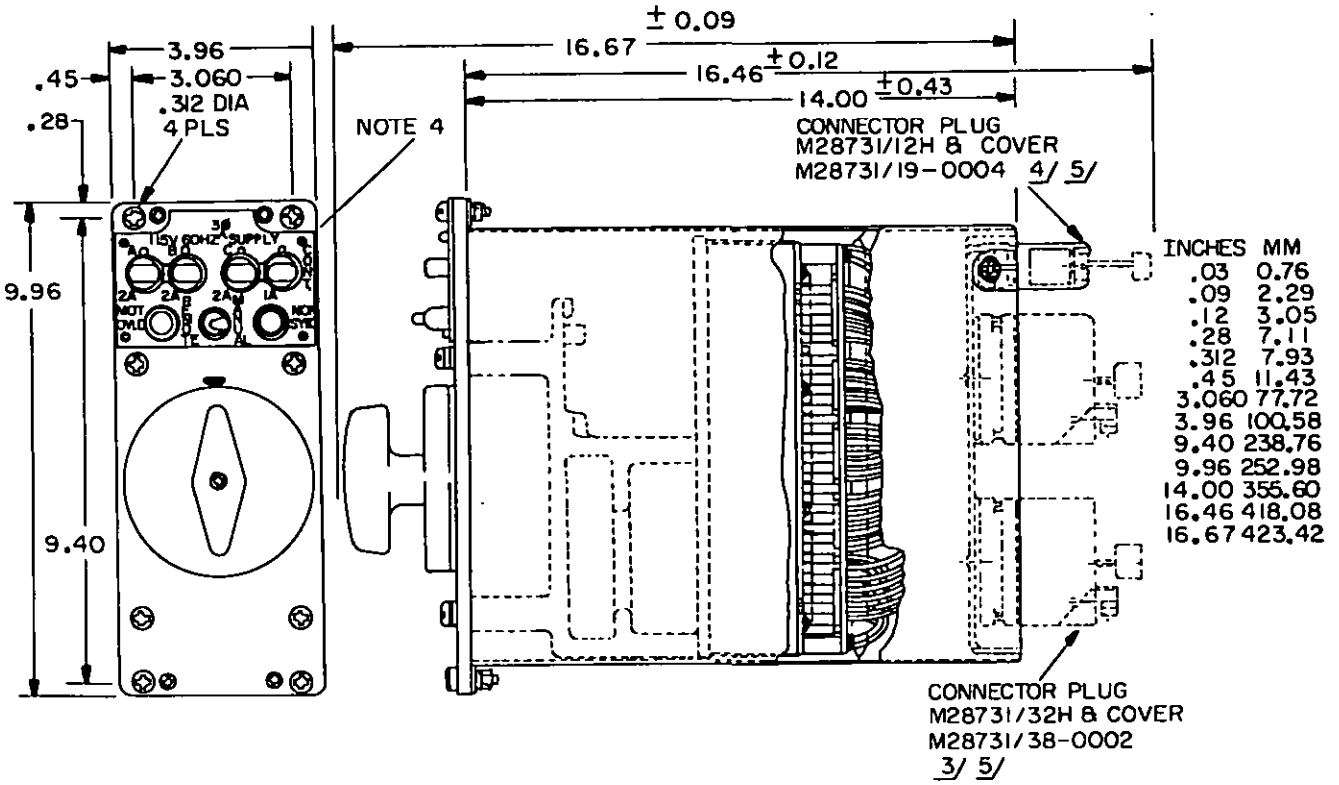


FIGURE 1. *Style 5BLS switch assembly.*

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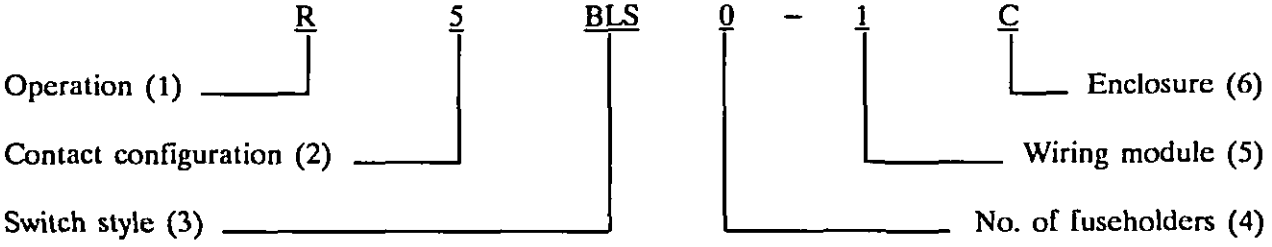
NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are plus or minus 0.02 (0.51 mm) for two-place decimals and plus or minus 0.005 (0.13 mm) for three-place decimals.
3. Connector cover, M28731/38-0002, may be rotated 180 degrees to accept cable from bottom.
4. Supplied on remote BLS only.
5. Mating connectors, supplied only when specified.

Metric equivalents (to the nearest 0.01 mm) are given for general information and are based upon inch = 25.4 mm.

FIGURE 1. *Style 5BLS switch assembly – Continued.*

Type designation:



(1) Switch assemblies shall be remote operated with manual override, designated R, or manually operated, designated M.

(2) Contact configuration shall be type 5 (see figures 2 and 3).

(3) Switch style shall be designated by the letters BLS.

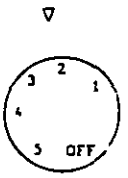
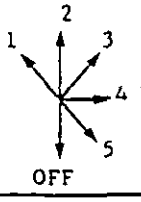
(4) There shall be no provisions for additional fuseholders.

(5) Wiring module basic circuitry shall be indicated by the designator 1 as specified to accomplish the switching actions specified in table I.

(6) Enclosed switch assemblies shall be indicated by a single letter C.

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TABLE I. *Switch selection.*

Type designation						Application data						
Switch operation	Contact configuration	Switch style	Additional fuscholders	Wiring module	Enclosure	*Switching action*[1]	*Stop position*	*Dial position*[2]	*Handle position*[3]	Wired contacts[4]	To ship connector[5]	Unwired contact[4]
M/R	5	BLS	0	1	C	1,2,3,4, 5, OFF	1&7			( )E LOAD 5 ( )D LOAD 4 ( )C LOAD 3 ( )B LOAD 2 ( )A LOAD 1 ( )X INPUT	JG JF JE JD JC JB	( )F OFF

Example: Type designation for the standard remote operated switch assembly shown with enclosure is R5BLS0-1C; the manual version is M5BLS0-1C.

[1]Stops shall be adjustable to limit switch handle rotation to the positions indicated or to any consecutive positions within the maximum rotation indicated.

[2]Reference position of a style 5BLS switch is considered to be that in which the flat on the handle shaft is up. The mark ∇ and the dial positions shown are used to designate the reference position.

[3]Normal position of a style 5BLS switch is the position which coincides with normal circuit conditions in any application. Handle shall be adjustable to point upward. The handle and rotary description plate rotate.

[4]See figures 2 and 3 for complete description of contact identification and table VI for load switch wiring.

[5]See figure 5 and tables VI and VII for connector identification and arrangement.

REQUIREMENTS:

Construction: Style 5BLS switch parts, modules and assemblies shall be constructed in accordance with Drawing 815-5727187.

Characteristics: Characteristics shown in table II shall apply to the style 5BLS switch assembly.

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TABLE II. *Characteristics.*

Characteristics	Value
Total number of poles available	72
Wired load switching poles	72
Number of positions	2 to 6
Break-before-make	Yes
Make-before-break	No
Electrical rating and endurance loading:	
4 Amperes, 115 volts, ac, resistive	20,000 Operations
3 Amperes, 115 volts, ac, (0.3 p.f.), 60 Hz	20,000 Operations
3 Amperes, 115 volts, ac, (0.1 p.f.), 400 Hz	20,000 Operations
4 Amperes, 28 volts, dc, resistive	20,000 Operations
25 Microamperes, 1.5 Vdc	20,000 Operations
Remote operation:	
Transfer time:	
Minimum	0.030 second
Maximum	0.125 second
Actuating time:	
1 Position change	0.25 second
5 Position change (one direction)	1.50 seconds max.
Current drain:	
at 115V, 3Ø, 60 Hz	1 Ampere rms per phase max. (0.70 Amp., typ.)
Contact resistance:	
At 4 amperes	0.025 ohm
At 25 microamperes	0.05 ohm
Total circuit resistance:	
At 4 amperes	0.025 ohm
At 25 microamperes	0.05 ohm
Torque – shaft rotation maximum – Initial	40 pound-inches
– After shock	40 pound-inches
– Stop strength	150 pound-inches
Maximum weight:	
Enclosed with wiring module	20 pounds
Without enclosure and wiring module	140 pounds
Temperature and humidity (for rotational torque and endurance tests):	
Minimum	Plus 40 °F (plus 4 °C)
Maximum	Plus 149 °F (plus 65 °C) at 40 to 50 percent rh
Shock	MIL-S-901, grade A, class I, type C
Vibration	MIL-STD-167-1, type I, except omit variable frequency
Temperature rise	30 °C

Switch assembly modules: Style 5BLS manually operated switch assemblies shall consist of the following modules as detailed on Drawing 815-5727187.

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Switch assembly:

Mechanism module
Switch contact module
Wiring module
Enclosure

Remote-operated, style 5BLS switch assemblies shall have, in addition to the above,

Remote control module/Mechanism module (associated circuitry) control circuit module.

Switch function: Style 5BLS switch assemblies shall provided 72 6-position poles each consisting of 7 stationary contacts X, A, B, C, D, E, and F and a movable contact arranged as shown on figure 2. A stop shall prevent transfer of the movable contact into position X. Additional stops (movable) shall be provided which, when inserted as specified in table III, will limit movement to two through six consecutive positions. Stops shall be as detailed on Drawing 815-5727187 and shall be accessible from the front of the rotary description plate.

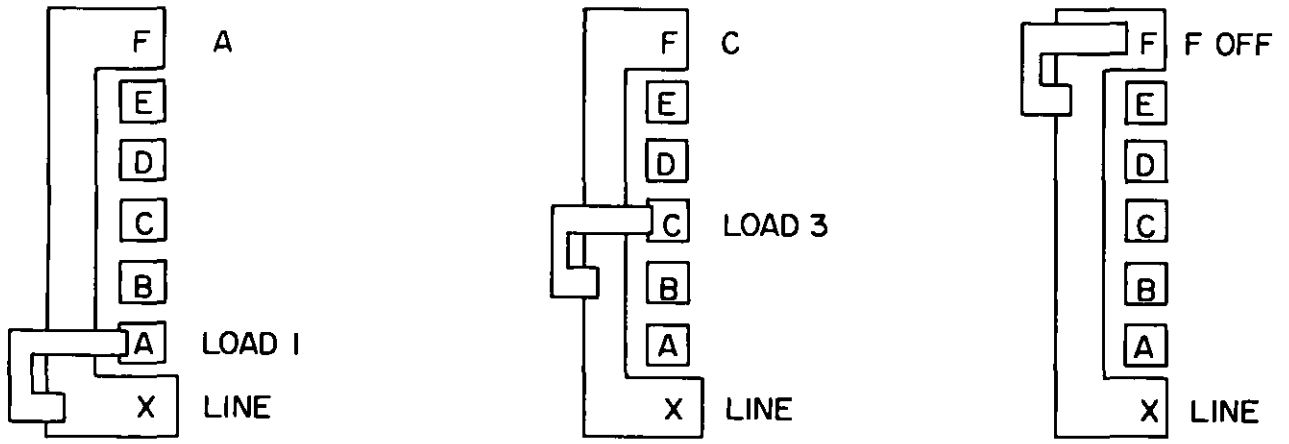


FIGURE 2. *Contact configuration.*

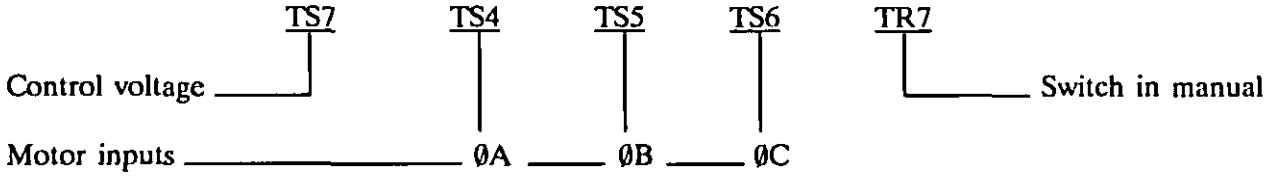
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TABLE III. *Stop positions versus switching action.*

Stop positions	Switch positions accessible	Switching action[1]
1 and 7	A-B-C-D-E-F	1-2-3-4-5-OFF
1 and 6	A-B-C-D-E	1-2-3-4-5
1 and 5	A-B-C-D	1-2-3-4
1 and 4	A-B-C	1-2-3
1 and 3	A-B	1-2

[1]With stops located as specified, a switch assembly shall provide from two to five active positions, any of which can be used as an OFF position by omitting external connections to the contacts in the desired OFF positions.

Remote operation: Connections to the switch module for remote operation shall be made to contacts as follows: (see table VII)



Remotely operated switch assembly shall move to the positions shown in table IV when the specified control voltage is applied.

TABLE IV. *Control voltage for remote operation.*

Control voltage	Closed switch contacts	XFMR tap[1]
0	X-A	A
16.4	X-B	B
32.8	X-C	C
49.2	X-D	D
65.6	X-E	E
82.0	X-F	F

[1]See figure 3 of MIL-S-24187 for typical test circuits for remote operated switch assemblies. Test loading is shown on figure 7 of this document.

[2]See figure 2 for typical pole positions corresponding to the transformer tap voltages A through F.

Table V lists the required connections in the wiring module for the remote control circuitry using type B-20 multi-strand grey wire of MIL-W-16878.

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TABLE V. *Wiring connections for remote operations.*

Control wiring contact	Circuit jumper contact
BR7	AFO
BS7	AEO
BR8	ADO
BS8	ACO
BS9	ABO
BS10	AAO
BS11	AXO

Enclosure and wiring module: Enclosure shall be furnished with the wiring module to connect the switch contact module to the ship's wiring connector receptacles at the rear of the enclosure (see figure 5). The general arrangement of the wiring shall be as shown on figure 4. Interconnecting wiring shall be as shown in tables VI and VII.

Contact designation: Each stationary switching contact shall be identified by two letters and a number. The first letter, A, B, C, D, or E shall identify the pole group in which a contact is located and the second letter shall be assigned as shown on figure 2. The number shall be that of the column in which a contact is located (see figure 3). Each contact in the switch contact module connector plug shall be assigned the same identification as the contact in the switch contact module to which it connects. Contact F is common with contact X in all poles. Contacts installed are signified by +, no contacts are signified by O.

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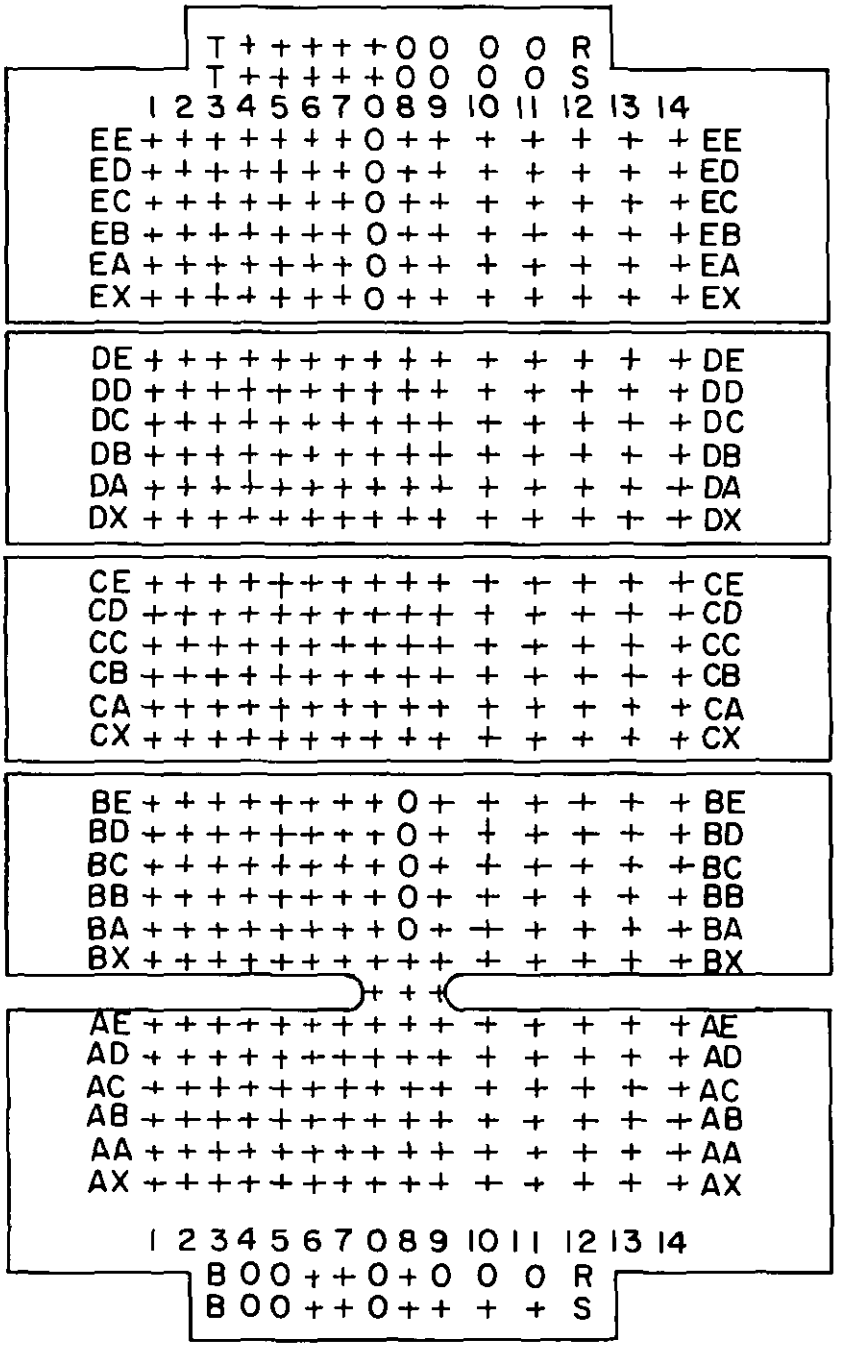


FIGURE 3. *Contact module marking, rear view.*

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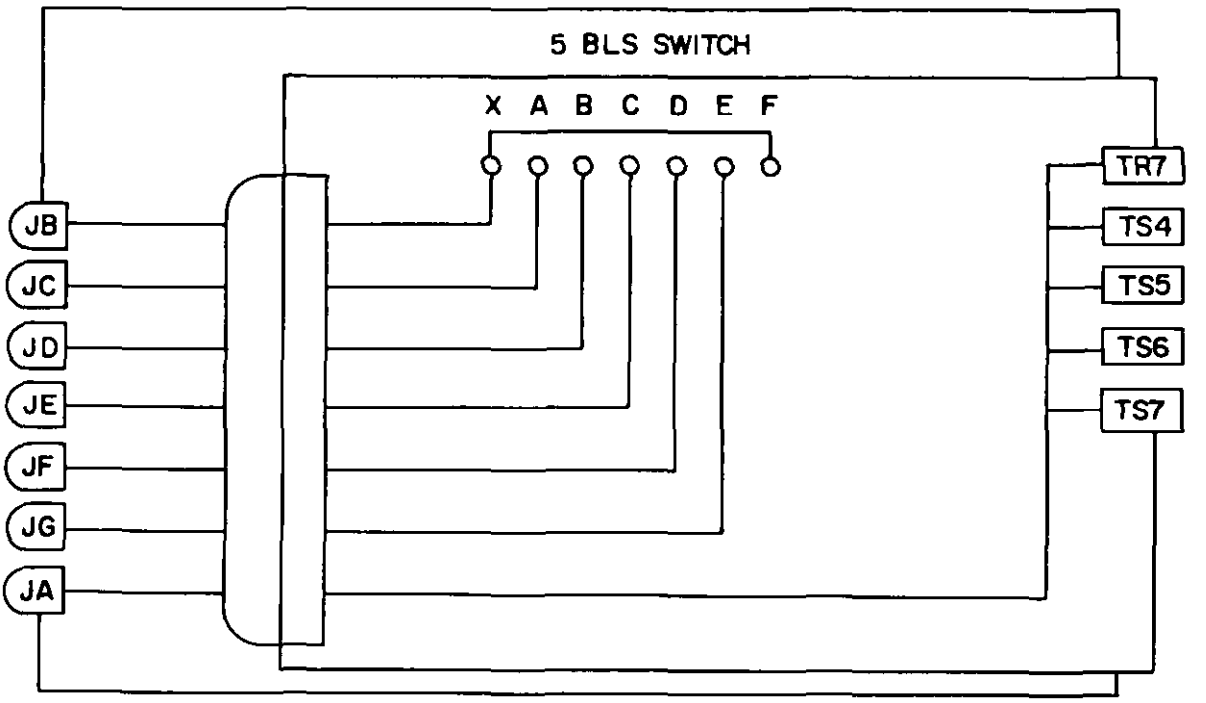


FIGURE 4. *Interconnecting wiring arrangement.*

Wiring module connections: For load switching circuits each 117 pin connector shall be wired to specific group of contacts of a switch contact module using the data shown in table VI and figure 5. Wire shall be type B-22, multistrand of MIL-W-16878. The color coding shall be as illustrated in the example below.

TABLE VI. *Contacts in switch contact module.*

	Switch contact module contact														
	Pole column														
Pole group	1	2	3	4	5	6	7	0	8	9	10	11	12	13	14
E	C2	B2	A2	A3	B3	C3	C4		A4	A5	B5	C5	C6	B6	A6
D	F2	E2	D2	D3	E3	F3	F4	E4	D4	D5	E5	F5	F6	E6	D6
C	N3	N2	M2	K2	J2	H2	J3	H4	J5	H6	J6	K6	M6	N6	
B	R2	S2	T2	T3	S3	R3	R4		T4	T5	S5	R5	R6	S6	T6
A	U2	V2	X2	X3	V3	U3	U4		X4	X5	V5	U5	U6	V6	X6
	117 Pin connector contacts														

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Example:

Each pole group contains switch contacts X, A, B, C, D, E, and F (see figure 4); contact F is unwired and is common to contact X.

Example:

Pole E5 contact X wired to pin B3 of connector JB black wire.
Pole E5 contact A wired to pin B3 of connector JC white wire.
Pole E5 contact B wired to pin B3 of connector JD red wire.
Pole E5 contact C wired to pin B3 of connector JE blue wire.
Pole E5 contact D wired to pin B3 of connector JF orange wire.
Pole E5 contact E wired to pin B3 of connector JG yellow wire.

The 38-pin connector contacts shall be wired to contacts in the switch contact module as shown in table VII.

TABLE VII. *Wiring module connections, 38-pin connector (JA).*

JA contact	Switch contact	Switch service
A	TR4	Not used
B	TS4	Phase A
E	TR5	Not used
F	TS5	Phase B
L	TR6	Not used
M	TS6	Phase C
R	TR7	SW in manual
S	TS7	Control
V	TRO	BFI Return
X	TSO	BFI Supply
JJ	CX14	
KK	CA14	Status
LL	CB14	
MM	CC14	Circuits
NN	CD14	
PP	CE14	

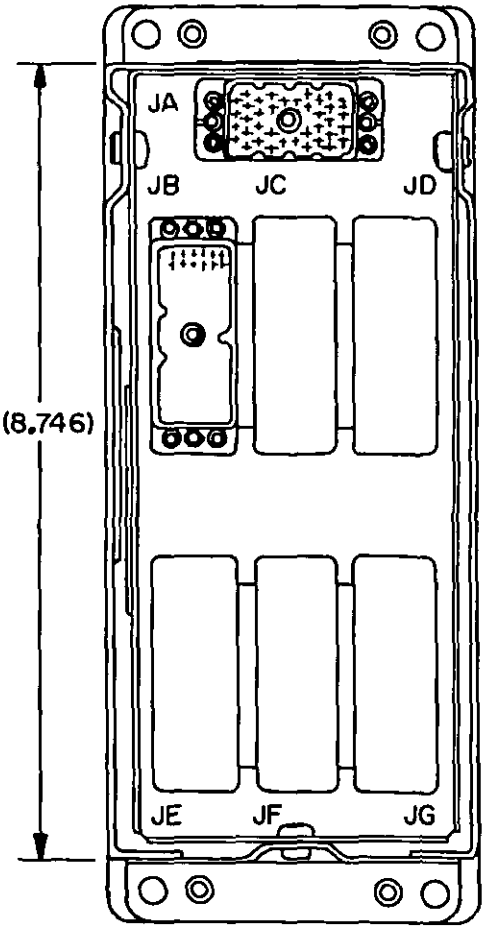


FIGURE 5. *Enclosed switch connector arrangements.*

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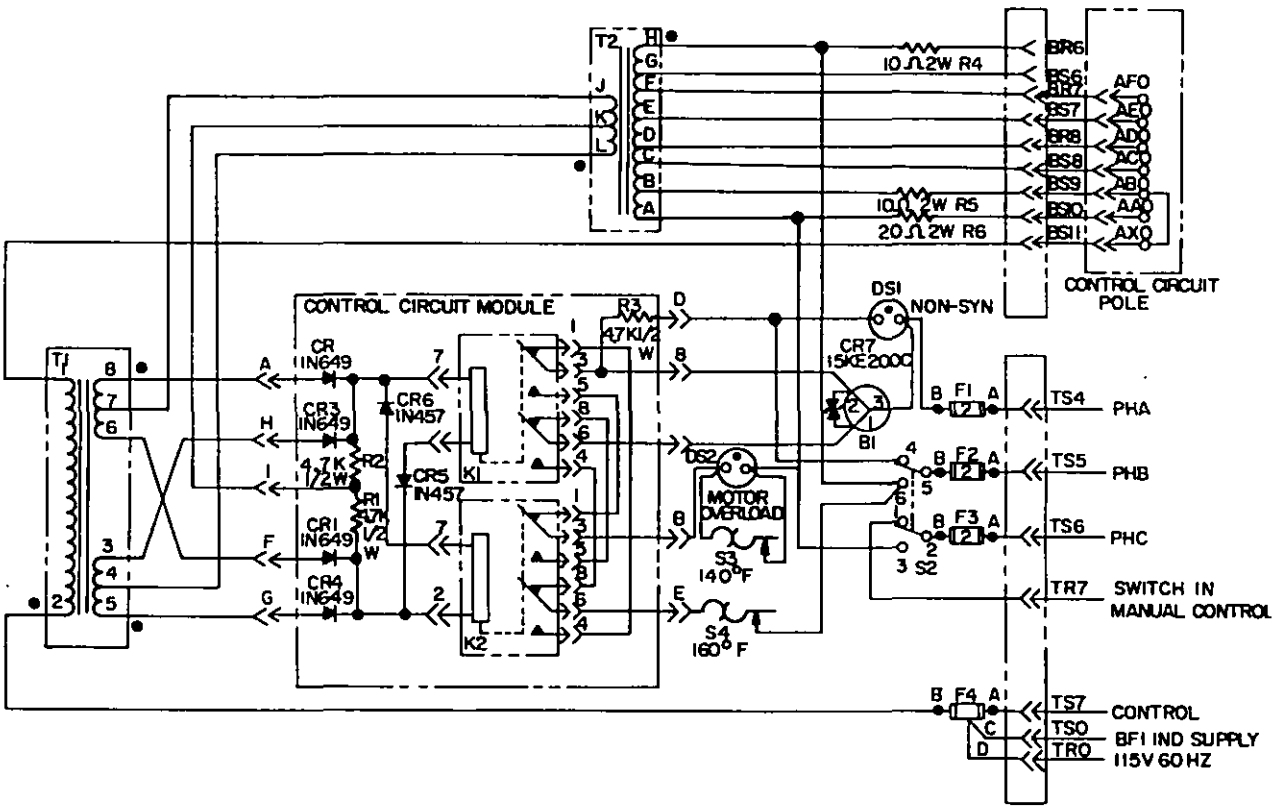


FIGURE 6. *Remote control circuit.*

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Repair parts: Repair parts shall be as specified in table VIII.

TABLE VIII. *Repair parts.*

Drawing 815-5727187 piece No.		Description	Quantities required for quantity of switches installed					
			1–3	4–12	13–30	31–60	61–120	121 or more
Manual	Remote							
2		Enclosed switch	1	1	1	1	2	2
	1	Enclosed switch	1	1	1	1	2	2
73	73	Stop screw	1	1	1	2	2	3
164	164	Detent spring	1	1	1	2	3	6
	87	Toggle switch	1	1	1	2	3	4
	85	Indicator light	1	1	1	2	3	5
	86	Indicator light	1	1	1	2	3	5
		Lamp (M15098/11-001)	Note 2					
	10	Control circuit module	1	1	1	2	2	3
	93	Relay	1	2	3	Note 3		
	160	Washer	2	2	2	4	6	8
	159	Spacer	2	2	2	4	6	8
	81	Fuseholder (less fuses)	1	1	2	3	4	5
		Fuse carrier (1A0981)	1	1	2	3	4	5
	82	Fuse, 2.0 ampere	Note 4					
	83	Fuse, 1.0 ampere	Note 4					

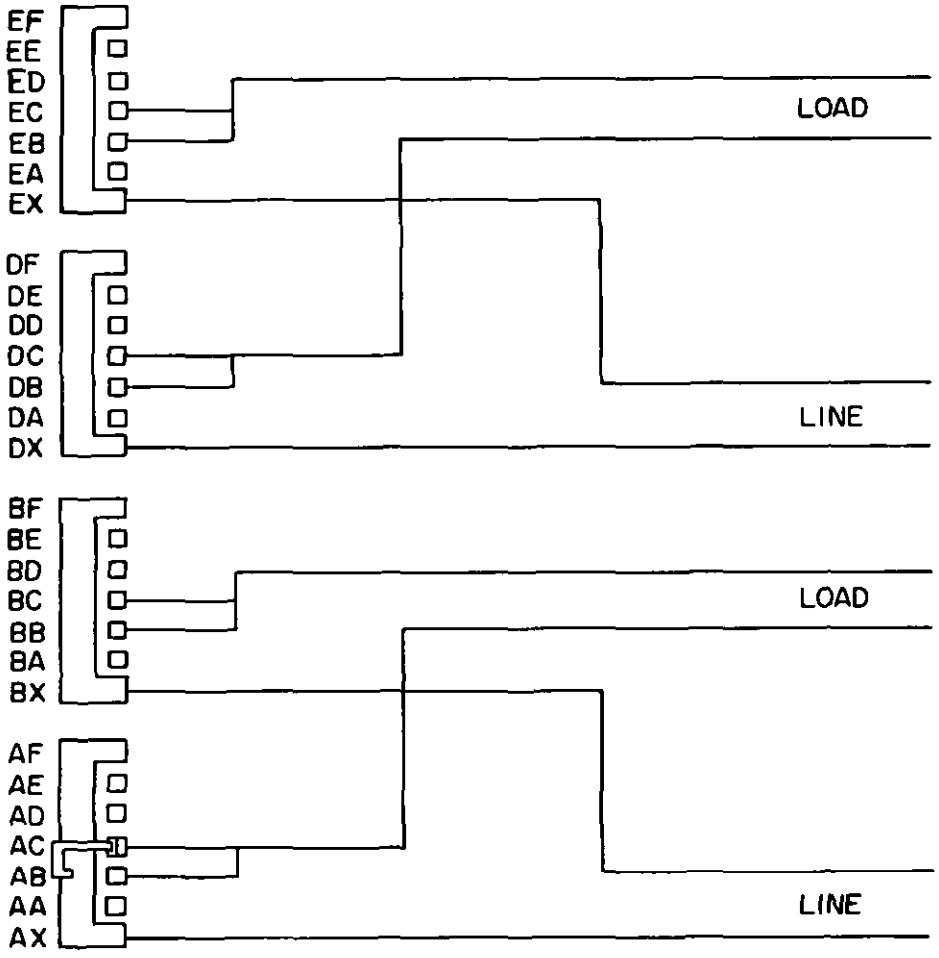
Notes:
1. Refer to Drawing 803-4680149 for piece numbers.
2. One hundred percent repair lamps shall be provided.
3. Supply one for each 10 or fraction of 10 installed.
4. One hundred fifty percent repair fuses shall be supplied. Fuses shall be supplied in multiples of five, *in standard commercial packages, with a minimum quantity of 15.*

Qualification samples: Two types R5BLS0-1C or type M5BLS0-1C switch assemblies shall be submitted for qualification inspection (RBLS qualifies MBLS; MBLS does not qualify RBLS).

Tests: Note: Manufacturers who have qualified to MIL-S-24187/2 are qualified by similarity.

Loading of switches: Loading of switches for endurance and temperature rise tests for qualification and quality conformance, group C testing shall be as shown on figure 7 and table IX.

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Column	Volts	Amperes	Power factor	Hz
1	115	4	1.0	60
4	115	3	0.3 Inductive	60
7	115	3	0.1 Inductive	400
11	28	4	Resistive	dc
14	1.5	25 μA	Resistive	dc

FIGURE 7. *Test loading.*

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Temperature rise: Contacts to be monitored during temperature rise test are shown in table IX.

TABLE IX. *Temperature measurements.*

Column	Contacts
1	AX, AB
1	DX, DB
1	EX, EB

Preparing activity:
Navy – SH
(Project 5930-N681-12)

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